

CU-423

RoboLink: Build and Drive Your Smartphone-Controlled Robot

Dr. Mahmoud **Sayed** & Mahmoud **Yassin**

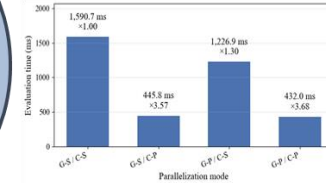
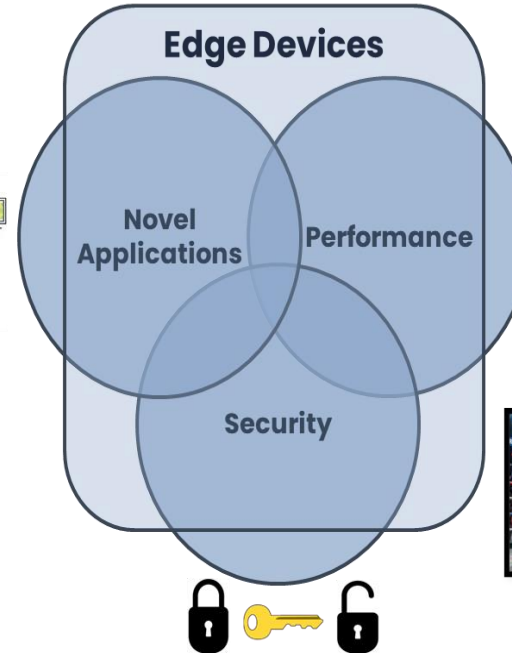
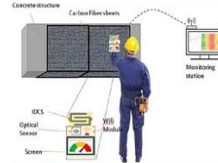
03-May-2026

Mahmoud Sayed?

- B.Sc, M.Sc., Ph.D in Electrical and Computer Eng.
- Postdoctoral Fellow – Sys. and Comp. Eng.
- What do I work in?

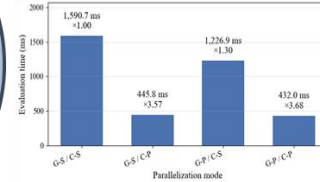
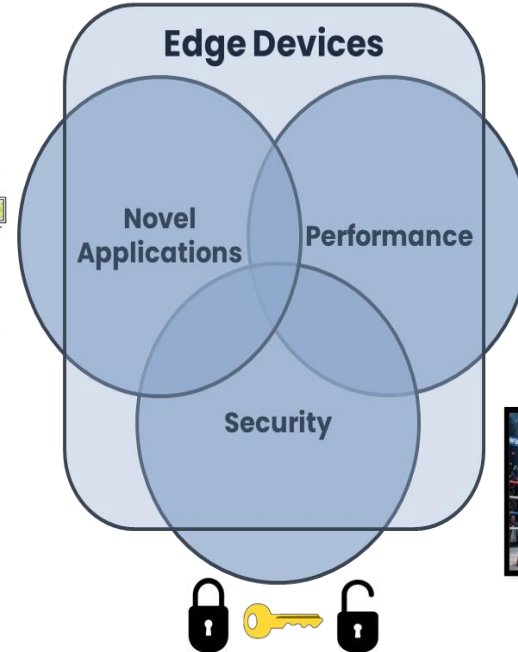
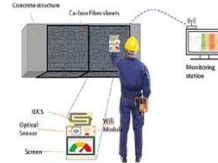
Mahmoud Sayed?

- B.Sc., M.Sc., Ph.D. in Elec. and Comp. Eng.
- Postdoctoral Fellow – Sys. and Comp. Eng.
- What do I do?



Mahmoud Sayed?

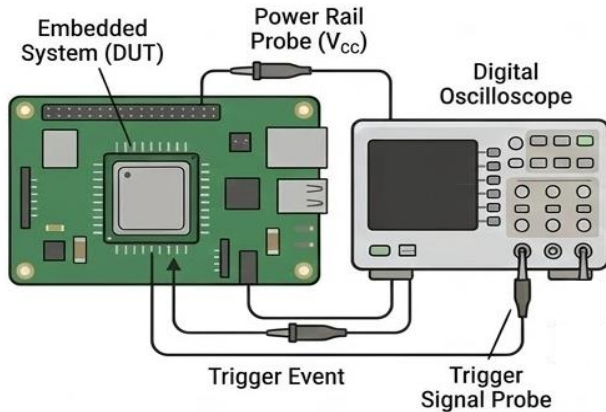
- B.Sc., M.Sc., Ph.D. in Elec. and Comp. Eng.
- Postdoctoral Fellow – Sys. and Comp. Eng.
- What do I do?



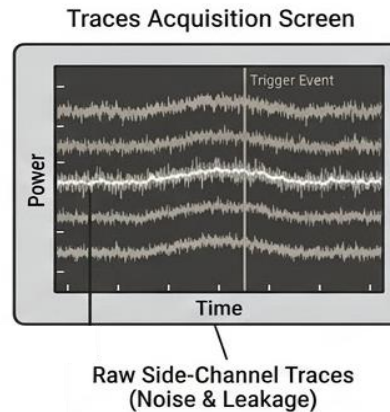
Mahmoud Sayed?

- What do I do? (Side Channel analysis)

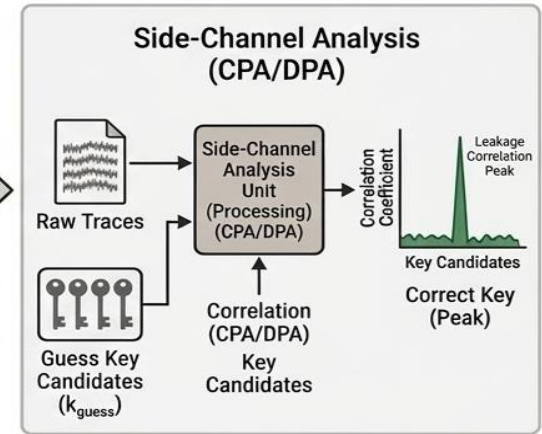
Part 1: Attack Setup
(The Connection)



Part 2: Signal Acquisition
(The Raw Traces)



Part 3: Side-Channel Analysis
(e.g., CPA/DPA)

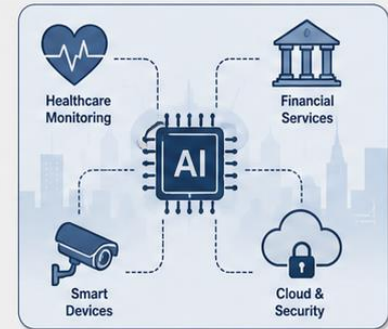
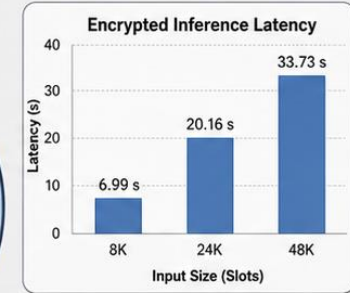
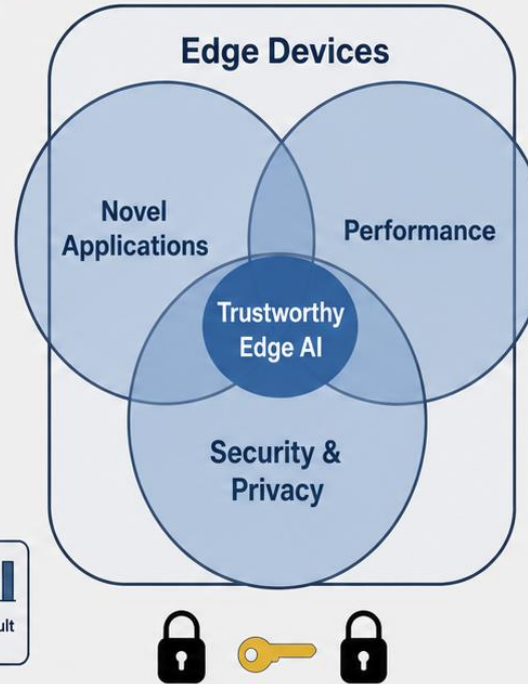
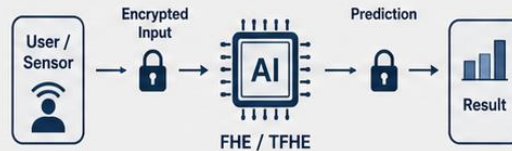


Mahmoud Yassin?

- B.Sc, M.Sc., Ph.D Candidate in Electrical and Computer Eng.
- Teaching Assistant – Sys. and Comp. Eng.
- What do I work in?

Mahmoud Yassin

- B.Eng. & M.A.Sc.; Ph.D. Candidate in Systems and Computer Engineering
- Research Assistant, Secure Cryptographic Implementations Lab
- Privacy-Preserving AI, Secure Edge AI, and Trustworthy Applications



What are we doing today?

Two Objectives



1-Housekeeping



2-What are we going to do in our Five days?

Housekeeping

Housekeeping

- Our room is SA509 (Southam Hall room 509)
- Attendance should be taken at 09:30 & 13:30
- Break time is from 11:30 to 13:00

Housekeeping

- We will work in groups
 - Because we have 7 kits.
- We will need to use the computers
 - Students accounts
- WiFi access:

Network: CU-Wireless
Domain: net.carleton.ca
Username: WifiW26
Password: Mac0drum!

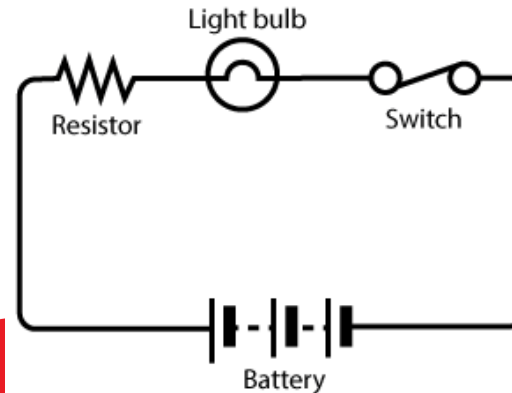
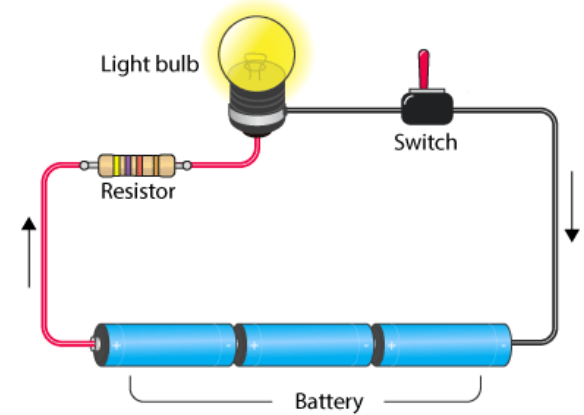
What are we going to do?

First, What do you expect to find?

Let me hear your thoughts!

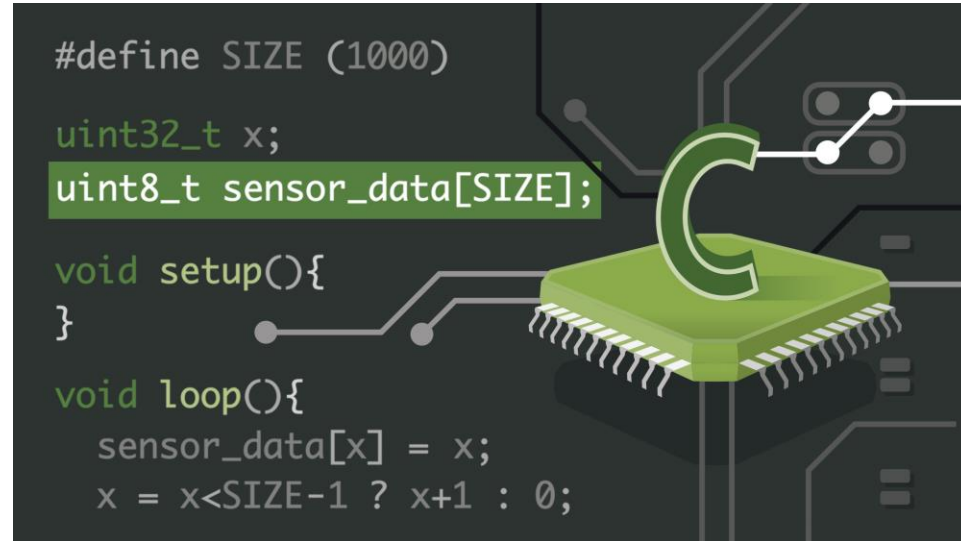
Day 1: Electricity and Circuits

- Learn about simple electric circuits.
 - Voltage, Current, Resistor
- Learn about LEDs, Switches.
- **Measure and see!**



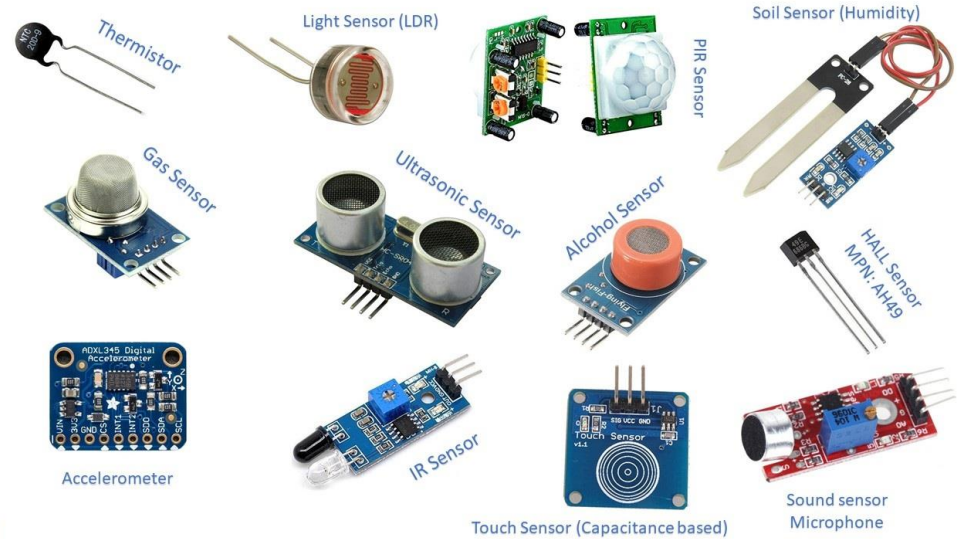
Day 2: Programming!

- Learn basic programming
 - C/C++
- **Write & run your first embedded program!**



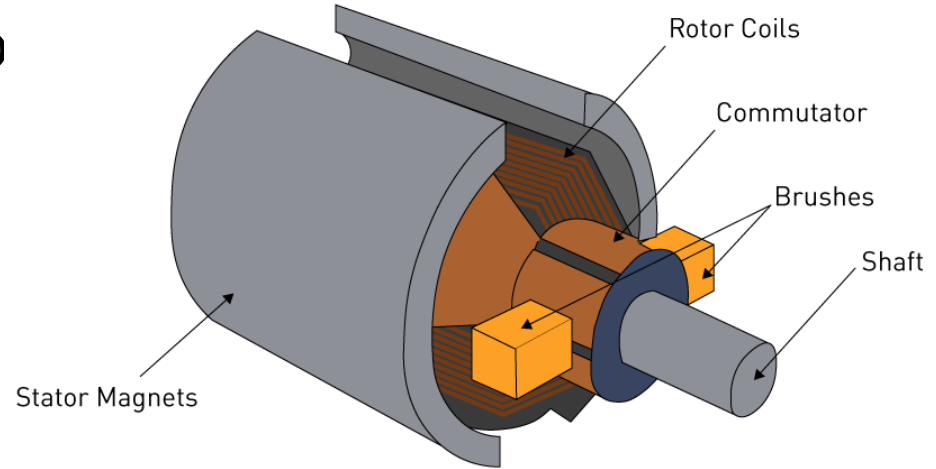
Day 3: Sensors!

- Learn sensing principles
 - How to sense object, distance, heat, ...!
- See ultrasonic sensor in action!



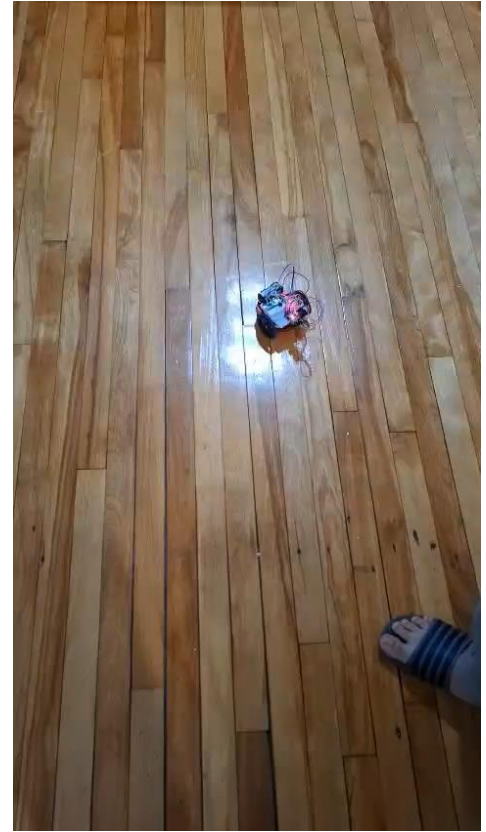
Day 4: Motors!

- Learn basic DC motor principle
 - How to power a motor?
 - How to steer a motor?
- **Your car's first run!**

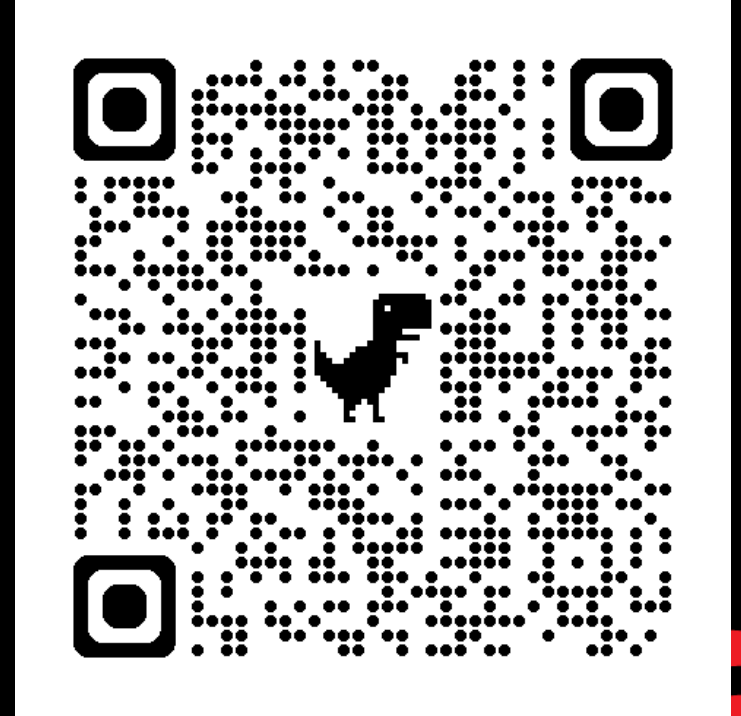


Day 5: THE PROJECT!

- Learn wireless communication.
 - How to control your robot wirelessly?
- **Drive your robot through your phone!**



A 2 questions survey!



Questions?

